Candidate Seat No:________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Bachelor of Computer Applications (BCA)
Semester-I University Examination March 2018
CA 114 - Foundation of Mathematics
Date: 29/03/2018, Thrusday   Time:10:00 a.m. to 01:00 p.m.   Maximum Marks: 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **The figure to the right indicates marks.**
3. **Make a suitable assumption when necessary.**

SECTION-I

Q.1 Do as directed.

1. A set which having a single element is known as equal set. **(True/False)** **[01]**
2. Identify the type of matrix. A=$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ **[01]**

   A. Identity Matrix   B. Singular Matrix   C. Zero Matrix   D. Column Matrix
3. $A \cap A'$ = ______. **[01]**
4. A matrix in which number of rows and columns are equal is said to be ________ matrix. **[01]**
5. f(x) = 3x + 3, D*f* = {5,2}. Find R*f.* **[02]**
6. Find AxB if A= {m,n} & B= {x,y,z} **[02]**
7. Find determinant value of A= $\begin{vmatrix}1 & 3 & 0\\ 1 & 2 & 1\\ 1 & 0 & 2\end{vmatrix}$ **[03]**

Q.2 [A] Find : B + A and A – B if, A = $\begin{bmatrix}1 & 3 & 0\\ 1 & 2 & 3\\ 1 & 3 & 4\end{bmatrix}$ & B = 2A **[05]**

[B] U = {a,b,c,d,e,f,g,h} A = { a,c,e,f,h} & B = { a,b,c,e,g} . Find $A' \cup B$ and $A \cap B'$. **[04]**

[C] $f(x)=3x^2 + 5$, g(x)=2x - 1 , where x={1,2}. Check whether f(x) = g(x). **[03]**

OR

Q.2 Define the given terms with example. **[12]**

- Disjoint Set
- Relation
- Zero Matrix
- One to Many Relation
- Tringular Matrix
- Union of two Set

Q.3 Answer following questions. (Any two) **[12]**

1. If g(x) = $\dfrac{x^2+2x}{x+2}$, find g(1) +g(3) + g(2)
2. Find inverse of A=$\begin{bmatrix}2 & 1 & 0\\ 5 & 1 & 0\\ 0 & 3 & 1\end{bmatrix}$
3. Find $A' - B$ and $A \Delta B'$, if U={1,2,3,4….10} A={1,3,5,6,7} & B= {1,2,5,6,8}
4. Find AB , if A = $\begin{bmatrix}5 & 3 & 0\\ 1 & 0 & 3\\ 1 & 3 & 4\end{bmatrix}$ & B = $\begin{bmatrix}2 & 3 & 0\\ 5 & 1 & 3\\ 0 & 3 & 2\end{bmatrix}$

SECTION-II

Q.**4** **Do as directed.**

1. Evaluate $\lim_{x\to 0} \frac{20x^2-15x+5}{15x^2-2x+1}$ **[01]**

2. If y = $\cos x$ then $\frac{dy}{dx}$ = ___________. **[01]**

3. $\int 5x^3 \ dx = \frac{5x^4}{4}$ + C. **(True/False)** **[01]**

4. If y = x^5 then $\frac{dy}{dx}$ = ________. **[01]**

5. The value of $\lim_{x\to 2} 5x + 3$ = ___________. **[01]**

6. Differentiation is the reverse process of Integration. **(True/False)** **[01]**

7. If y = e^x then $\frac{dy}{dx}$ = e^x. **(True/False)** **[01]**

8. Find Integration: $\int(\ e^{5x} \ + \ x^2)$ dx **[02]**

9. Find derivative: y = $2x^3 + \ 5x^2 + 3x$ **[02]**

Q.5 **[A]** Find derivative of y = $(3x^2+2)(2x+3)$ **[04]**

[B] Evaluate $\lim_{n\to\infty} \frac{15n^2-14n+1}{5n^2+5n-2}$ **[04]**

[C] Evaluate $\int_1^4 (5x^2 - 8x \ + \ 5)dx$ **[04]**

OR

Q.5 **[A]** State properties of derivatives. **[06]**

[B] Explain the following terms with example. **[06]**

(1) Differentiation **(2)** Integration

Q.6 **Answer following questions. (Any two)** **[12]**

1. Find second derivative of $f(x)$ = $4x^3 + 2x^2 + 5x + 2$ and also find $f\ (3)$ and $f''\ (2)$.

2. Examine the continuity of the function defined by

$$f(x) = \begin{cases} x^2, \ x \le 0 \\ 5x-4, \ 0 < x \le 1 \\ 4x^2-3, \ 1 < x < 2 \end{cases} \qquad At\ point\ x = 0{,}1$$

3. Evaluate $\lim_{x\to 3} \frac{x^3-4x^2-3x+18}{x^3-8x^2+21x-18}$

4. Find the area bounded by the parabola y = x^2+x+4, when the y-axis on the line x = 1 and x = 3.